Tails + significance + rejection regions

A1S2: SLIDES 1-51 ONLY

1 RECOGNISE IT

A1S2 SCOPE:
Tail recognition, H_1 direction, α placement and rejection-region recognition are in scope.
The later detailed one-tail mean-test workflow after slide 51 is excluded from the current A1S2 Chapter 9 scope.

2 DECISION / METHOD

- H_1: $\mu \neq \mu_0$ → two tails, α/2 each.
- H_1: $\mu < \mu_0$ → left tail, α.
- H_1: $\mu > \mu_0$ → right tail, α.

Place the boundary/boundaries and recognise the rejection region; do not extend into later workflows.

3 FORMULA

$z^* = \Phi^{-1}(1-\alpha/2) > 0$
two-tail criticals = $\pm z^*$
left critical = $\Phi^{-1}(\alpha)$
right critical = $\Phi^{-1}(1-\alpha)$
reject H_0 only inside rejection region

4 FORMULA SHEET VS WHAT I KNOW

Gives: no critical-value map.
Still know: H_1 controls tails; significance is total rejection-region area under H_0.

5 EXCEL ROUTE

- Two-tail: `=NORM.S.INV(alpha/2)` and `=NORM.S.INV(1-alpha/2)`
- Recognition only: left NORM.S.INV(α); right NORM.S.INV(1-α).

6 TINY WORKED EXAMPLE

If H_1:$\mu<20$ and $\alpha=.05$, it is a left-tail rejection region with Z critical about -1.6449.

7 TOP EXAM TRAPS

- Splitting α incorrectly between rejection regions.
- Choosing tail from sign of zcalc rather than H_1.
- Saying 'accept H_0'.
- Using the wrong critical-value sign.

8 SAY ALOUD

The alternative hypothesis chooses the tail; α is the total rejection-region area.

9 EXIT CHECK

Which H_1 creates two rejection regions?

ONE-GLANCE VISUAL ROUTE

